Final Scheme of Valuation/Answer Key

(Scheme of evaluation (marks in brackets) and answers of problems/key)

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

THIRD SEMESTER M.C.A. DEGREE EXAMINATION, DECEMBER 2021

Course Code: 20MCA203

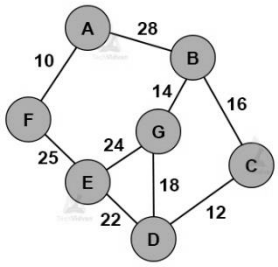
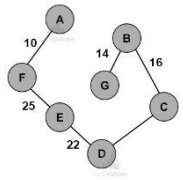
Course Name: DESIGN & ANALYSIS OF ALGORITHMS

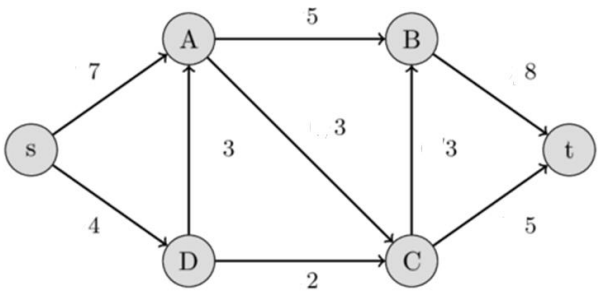
Max. Marks: 60		Duration: 3 Hours

PART A

		Answer all questions, each carries3 marks.	Marks
1		Differentiate space and time complexity. **Keys** Explanation of space complexity (1.5 marks) and time complexity (1.5 marks)	(3)
2		Solve the following recurrence equation using substitution method T(n)=2T (n/2) + n **Keys** Time complexity = O (n log n)	(3)
3		Define the control abstraction of Greedy strategy **Keys** Define greedy strategy (1 mark) ,Control abstraction of greedy strategy(2 marks)	(3)
4		Explain the pseudocode of Bellman ford algorithm for finding the single shortest path **Keys** Pseudo code (1.5 marks) Explanation (1.5 marks)	(3)
5		Describe the working of Backtracking problem with a suitable example **Keys** Backtracking definition (1 mark) Working of Backtracking with an example (2 marks)	(3)
6		Write a comparison-based sorting algorithm used by lower bound theory technique **Keys** Definition of lower bound theory (1 mark)	(3)

		Example: - sorting of n numbers has a lower bound (2 marks)	
7		Explain Clique problem with an example **Keys** Explanation – 2 marks Example: - 1 mark	(3)
8		Explain the Bipartite matching problem with an example **Keys** Explanation of Bipartite matching (2 mark) Example. (1 marks)	(3)
9		Describe approximation ratio in approximation algorithm. **Keys** Definition – (2 mark) Equation – (1 marks)	(3)
10		What are the two different types of Randomized algorithms **Keys** Lasvegas (1.5) and Montecarlo algorithms ((1.5)	(3)
PART B			
Answer any one question from each module. Each question carries 6 marks.			
Module I			
11		Describe different Asymptotic notations used for expressing Time complexity **Keys** Big O(2) , Big Omega(1), Theta(1), Little o(1), liitle Omega(1)	(6)
OR			
12		Explain the Merge sort algorithm and give its worst-case analysis **Keys** Explanation of Merge sort algorithm (3) worst-case analysis (3) marks	(6)
Module II			
13		Explain the different steps used to find the minimum cost spanning tree for the below graph using Prim's algorithm	(6)

		 **Keys** The final obtained minimal spanning tree is:  Cost of MST calculated as =10+25+22+12+16+14=99	
		OR	
14		Give the different steps for finding the shortest path in the all-pair shortest path algorithm with a suitable example. **Keys** Defining the steps (4 marks), Example (2 marks)	(6)
		Module III	
15		Discuss the sum of subsets algorithm with the following example A = {1,2,5,6,8}, Find the all-possible combinations whose sum is equal to the given value M = 9 **Keys** Defining the sum of subsets algorithm (3 marks), Problem solving (3 marks)	(6)
		OR	
16		Explain the Branch and Bound algorithm used for solving a 8-puzzle problem **Keys** Defining Branch and Bound algorithm (3), 8-puzzle problem solution (3)	(6)
		Module IV	
17		Prove that Vector cover problem is NP Complete. **Keys** P , NP, NP hard, NP Complete (3) , Vector problem solution (3)	(6)
		OR	

18		Compute the maximum flow in the below network using Ford- Fulkerson algorithm  **Keys** Ford – Fulkerson Algorithm – (3) Computation of maximum flow (Answer : maximum flow=10) (3 marks)	(6)
		Module V	
19		Explain the 2-Approximation algorithm for vertex cover and justify its approximation ratio **Keys** Defining Approximation algorithm (3) Vertex cover problem solution –(3)	(6)
		OR	
20		What is Randomised Quick sort? Explain its working and why is it used **Keys** Defining Randomised Quick sort (3) Explain its working and why is it used (3)	(6)
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